
Voltage Reducer

DESCRIPTION AND RECOMMENDATIONS

This device is intended for reducing DC voltages by a proportion, to supply non-changing loads. It generates PWM from a DC supply. Closing the jumper connects an internal RC circuit that attempts to decouple the voltage. A large decoupling capacitor at the output is recommended; 470 μ F is found to reduce the voltage tolerance below $\pm 10\%$.

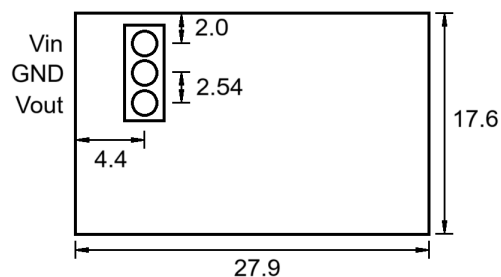
NOTES

All values are in the context of $T_a = 25^\circ\text{C}$.

Some values are presented in the form “*measured (theoretical)*”.

PIN CONFIGURATION AND DIMENSIONS

All dimensions are in millimeters.



MAXIMUM RATINGS

Parameter	Symbol	Minimum Value	Maximum Value	Unit
Maximum Input Voltage	V_{CC}	0	18	V
Output Current	I_o	-100	100	mA

OUTPUT VOLTAGE CONTROL

Parameter	Symbol	Minimum Value	Maximum Value	Unit
Output Percentage of V_{CC}	V_o/V_{CC}	0 (0.990)	79.8 (100)	%
Output Voltage Tolerance	$\Delta V_o/V_o$	± 21.1	± 25.0	%
Power Efficiency	P_o/P_{in}	0.0000131	16.8	%

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PWM CONTROL

Parameter	Symbol	Minimum Value	Maximum Value	Unit
On Time	t_{on}	1.30 (0.693)		μs
Off Time	t_{off}	0.280 (0)	134 (69.3)	μs
Duty Cycle	$t_{on}/(t_{on} + t_{off})$	0.961 (0.990)	82.3 (100)	%
Frequency	f	6.40 (14.3)	155 (1440)	kHz

INPUT CURRENT (mA)

Measurements were taken at $V_{CC} = 9V$.

Load Resistance (Ω)	Output Voltage (V)				
	7	5	3	1	0.1
No Load	29.9	126	149	117	85.7
100k	41.2	148	151	117	85.0
10k	38.2	138	151	116	86.4
1k	43.6	142	154	118	85.6
100	-	165	172	123	-

OUTPUT VOLTAGE TOLERANCE, PEAK TO PEAK (V)

Measurements were taken at $V_{CC} = 9V$.

Load Resistance (Ω)	Output Voltage (V)				
	7	5	3	1	0.1
No Load	3.8	4.0	3.9	4.3	4.5
100k	3.8	4.0	3.9	4.3	4.4
10k	3.8	4.1	3.9	4.2	4.3
1k	3.8	4.0	3.9	4.3	4.3
100	-	3.8	3.8	3.9	-

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POWER EFFICIENCY (%)

Values were calculated from the [Input Current](#) table.

Load Resistance (Ω)	Output Voltage (V)				
	7	5	3	1	0.1
No Load	0	0	0	0	0
100k	0.132	0.0188	0.00662	0.000950	0.0000131
10k	1.43	0.201	0.0662	0.00958	0.000129
1k	12.5	1.96	0.649	0.0942	0.00130
100	-	16.8	5.81	0.903	-

POWER EFFICIENCY VERSUS OUTPUT CURRENT

